

0. Day 1 (retroactive) — Risk & Vulnerability Data

Availability Note

Confirmed source: Ahmedabad Municipal Corporation's own website (ahmedabadcity.gov.in/SP/Demographics) publishes a direct download for **ward-wise Census 2011 data** for the AMC area. City-level summary confirms it includes: total population, male/female population, 0–6 age population, SC/ST population, literacy — all broken down by ward.

Feature-by-feature availability (best current estimate — Garima to verify exact fields when she downloads the file):

Feature	Status	Notes
Population / density	● Likely available	Ward-wise population confirmed; density derivable once ward area is known from the boundary file
SC/ST population	● Available	Can serve as a partial socioeconomic vulnerability proxy if needed
Elderly % (60+)	● Needs verification	City summary only shows a 0–6 age breakdown; the full ward-wise file needs to be checked for a 60+ or full age-group breakdown
Outdoor worker density	● Unlikely at ward level	Rarely published this granularly; if unavailable, drop this feature rather than invent a number — 3 real features beat 4 with one guessed
Informal settlement / slum density	● City-level only so far	City-wide slum population figure exists (~440,000 people); ward-level breakdown needs a separate check (AMC/Gujarat Slum Rehabilitation data)
Healthcare access	● Needs separate source	Not part of the Census file — would need a list of AMC hospitals/health centers, geocoded to wards
Green cover / NDVI	● Always available	Computed independently from Sentinel-2 satellite imagery, doesn't depend on any city portal

Action for Garima (Day 1/3): download the actual ward-wise file from the link above and confirm which of the ● items are genuinely present before Day 3's spatial join. If elderly %

isn't in the file, use SC/ST % or literacy as an alternate socioeconomic vulnerability proxy — but document that substitution honestly in the Decision Log rather than presenting it as elderly data.

Window chosen: 06:00 – 18:00 (12 hours)

Formula: $\text{DayStress} = \text{mean}(\text{UTCI})$ across daytime hours, per date

Why mean, not max: a single extreme hour shouldn't dominate the day's score — mean gives a more stable, representative daily thermal stress value.

Result (calculated on real ERA5 data, 2021–2026):

- 546 days processed
- DayStress range: 21.3°C to 43.8°C UTCI
- Output file: `daystress.csv` (date, DayStress)

2. Nighttime Window & Baseline

Window chosen: 20:00 – 06:00 (10 hours, wraps past midnight)

Baseline method: 90th percentile of all nighttime UTCI values, calculated across the **full 6-year dataset** (2021–2026) for Ahmedabad.

Why the full 6-year dataset: a baseline built from just one week would be arbitrary and easy for a judge to challenge. Using 6 years of real nighttime observations makes it a genuinely defensible climatological reference point, not a guess.

Result (calculated on real data):

- 49,140 nighttime hourly readings used
- Nighttime UTCI mean: 29.34°C
- Nighttime UTCI median: 29.36°C
- **BASELINE (90th percentile) = 36.34°C UTCI**

This is now a **frozen constant** for the NCTL formula:

$$\text{NCTL}(t) = \sum \max(0, \text{UTCI}_{\text{night}}(h) - 36.34) \times \Delta h$$

3. Risk Rubric v1.0 (Final Draft)

A three-dimensional priority matrix combining Thermal Burden (CTBI), Vulnerability, and Response Capacity.

Thermal Burden (CTBI)	Vulnerability	Response Capacity	Heat-Health Priority	Recommended Action Level
Low	Any	Any	Monitor	Routine surveillance
Moderate	Low	Any	Moderate	Public advisories
Moderate	High	High	High	Targeted interventions
Moderate	High	Low	High	Targeted interventions
High	Low	High	High	Enhanced monitoring
High	Low	Low	Very High	Preemptive resource deployment
High	High	High	Very High	Activate cooling centers + welfare checks
High	High	Low	Critical	Emergency response + hospital surge prep

Operational definitions (so "High/Low" isn't arbitrary):

- **Thermal Burden (CTBI):**
 - Low: CTBI < 50th percentile of historical distribution
 - Moderate: 50th ≤ CTBI < 85th percentile
 - High: CTBI ≥ 85th percentile
 - *(Exact percentile cutoffs get calibrated after Day 3, once Dev's real CTBI distribution exists — this rule stays fixed, only the numeric thresholds get filled in.)*
- **Vulnerability = High** if ≥2 of the following are above the city median:
 - Elderly population (%)
 - Population density
 - Outdoor worker density
 - Informal settlement density
- **Response Capacity = Low** if BOTH are below the city median:

- Healthcare access (hospital beds per capita / distance to primary health centers)
- Green cover (ward-level NDVI)

Why this structure: it follows the same logic used by the Heat Resilience Framework applied to Mumbai, Thane, and Nagpur — evaluating exposure, sensitivity, and response capacity as separate dimensions rather than one blended score, because identical heat exposure doesn't create identical risk across wards with different demographics and infrastructure. The graduated priority levels (Moderate → High → Very High → Critical) also match the general WHO/WMO approach of tying heat-health response to escalating alert tiers rather than a single on/off threshold.

Decision Log note: this rubric is explicitly labeled  Prototype — transparent, rule-based, and grounded in established frameworks, but not derived from Ahmedabad's own real health-outcome data (which we don't have access to as students). If asked "why these specific weights/thresholds," the honest answer is: adapted from established heat-vulnerability assessment practice, to be calibrated with real data in a future phase.

Handoff

- `daystress.csv` and the baseline value (36.34) go to **Dev**, so he can plug them directly into the CTBI formula on his chosen single continuous heatwave window.
- This document goes to the **Leader**, for the Decision Log and Day 3–4 PPT slides.